Kunal Prajapati

410210 | (91)8054449441| kunal444463@gmail.com | LinkedIn | Portfolio

SUMMARY

Radiation Physicist with expertise in radiation dosimetry, safety compliance, and particle accelerator-based systems, including proton therapy. Experienced in radiation field measurements, personnel dosimetry, and regulatory compliance in high-energy environments. Proven ability to implement radiation protection programs and ensure safe operation of complex radiation facilities.

EDUCATION

Homi Bhabha National Institute, BARC — Aug 2020
Post MSc. Diploma in Radiological Physics — 85.17%

Indian Institute of Technology, Delhi — May 2019
Master of Science (MS) in Physics — CGPA 7.725

Panjab University — May 2016
Bachelor of Science (BS) — 78.55%

CERTIFICATIONS

- Radiation safety officer (RSO), level III, *Atomic Energy Regulatory Board, India* — Dec 2022
- Accelerator Physics, *National Program on Technology Enhanced Learning (NPTEL)* — Oct 2022
- Deep Learning with PyTorch for Medical Image Analysis, *Udemy* — Jun 2026
- CS50's Introduction to Programming with Python, *Harvard University* — Dec 2024
- Python for Data Science, *National Program on Technology Enhanced Learning (NPTEL)* — Aug 2022
- Good clinical practice, *NIDA Clinical Trials Network* — Jan 2023

SKILLS

- **Technical:**

Machine learning	Python/Scikit-learn	Data Visualization	HTML/CSS
MATLAB (beginner)	Origin	SQL	3D Printing (CAD)

- **Languages:**

English (Fluent)	Hindi (Native)	Punjabi (Fluent)

WORK EXPERIENCE

IBA Proton Therapy, TMC Mumbai — Navi Mumbai
Beam Physics Service Engineer & Radiation Safety Officer — May 2023 – Present

- Maintained and troubleshot proton therapy accelerator and beam delivery systems, ensuring reliable clinical operation and high system availability.
- Contributed to the installation and commissioning of Proteus® PLUS proton therapy systems in Shenzhen and Chengdu, China, performing beam optics tuning and nozzle calibration.
- Conducted radiation surveys, personnel dosimetry assessments, radiation safety audits, and radioactive waste management to support safe clinical and technical operations.
- Provided technical support during machine QA, patient treatments, and clinical operations, while performing preventive and corrective maintenance on hardware and software systems.

Advanced Centre for Treatment, Research, and Education in Cancer (ACTREC) Navi Mumbai
Medical Physicist Feb 2022 – May 2023

- Conducted treatment planning and QA for advanced radiotherapy techniques like VMAT, TBI, TSET, TMLI.
- Contributed to clinical research in Medical Physics and Radiation Oncology.
- Performed HDR Brachytherapy planning and QA.

Tata Memorial Hospital, Mumbai Mumbai
Resident Medical Physicist (internship) Jan 2021 – Dec 2021

- Conducted treatment planning and QA for different radiotherapy techniques.
- Carried out HDR brachytherapy planning and QA.

HANDS-ON EXPERIENCE WITH RADIOTHERAPY SYSTEMS

- **Proton Therapy System**
  IBA Proteus PLUS

- **LINACs**
  Varian Truebeam Varian Novalis Tx Varian Trilogy Accuray Tomotherapy

- **Brachytherapy Units**
  Elekta microSelectron V3 afterloading system Elekta Flexitron afterloading system

PUBLICATIONS

- **Sigmoid dose accumulation and reporting for multifractionated brachytherapy for cervical cancer**
  Bindal, M. A., Mittal, P., Shinghal, A., Scaria, L., **Prajapati, K**., et al. (2023)

- **Spatially fractionated radiotherapy using straight and bending virtual rods as an alternative for cervical cancer brachytherapy—results from preclinical in-silico dose comparison and quality assurance**
  Shinde, P., A. K., Agarwal, P., Kamalnath, J., Joshi, K., Prajapati, K., et al. (2026)

- **Patterns of relapse after adjuvant (chemo)radiation for cervical cancer in a phase III clinical trial (PARCER): an evaluation of updated NRG Oncology /RTOG target delineation guidelines**
  Mittal, P., Chopra, S., Charnalia, M., Dora, T., Engineer, R., **Prajapati, K.**, et al. (2022)

ABSTRACTS

- **Validation of sigmoid points for sigmoid dose accumulation and reporting for multifractionated brachytherapy for cervical cancer**
  Kamalnath, J., A. K., **Prajapati, K.**, et al. (2024)

- **Sigmoid Points predict Late Toxicity Assessment for Multifractionated Brachytherapy for Cervical Cancer**
  Mittal P., Kamalnath, J., A. K., **Prajapati, K.**, et al. (2024)

PROFESSIONAL AFFILIATIONS

Association of Medical Physicists in India (AMPI), *Medical Physicist* Life-time member